Advanced Python Programming

Project Write-up

EDUPULSE

A Personalized E-Learning Platform with AI-Based Quiz Generation and Learning Resource Recommendations

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Course	:	Advanced Python Programming
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1. Abstract

EduPulse is a personalized e-learning platform designed to provide students with an integrated learning experience. The platform combines course management, learning materials, quizzes, assessments, performance analysis and personalized learning-resource recommendations.

Students can access courses, study topics, attempt quizzes of different difficulty levels and evaluate their learning progress. The platform uses Google Gemini API for quiz question generation, YouTube Data API for retrieving educational videos and the StudyCompass API for personalized learning-resource recommendations.

EduPulse aims to simplify the learning process by connecting student performance with suitable learning resources, helping students identify areas that require improvement and continue learning effectively.

2. Introduction

Online learning platforms provide students with access to educational materials and digital resources. However, students often use different platforms for studying, practising questions, taking assessments and finding additional learning materials.

EduPulse addresses this issue by bringing these activities together in a single web-based platform. It provides a structured learning environment where students can access courses, attempt quizzes, monitor their performance and receive learning-resource recommendations based on their learning progress.

The platform focuses on making learning more organized, accessible and personalized.

3. Problem Statement

Students often face difficulties in managing learning materials, identifying weak topics and finding appropriate resources for additional practice.

Traditional learning platforms may provide course materials and assessments separately, requiring students to manually search for supplementary resources.

There is a need for an integrated platform that connects learning activities, assessment results and personalized resource recommendations to support continuous learning.

4. Objectives

- To develop a personalized web-based e-learning platform.
- To provide course-wise and topic-wise learning materials.
- To generate quizzes with different difficulty levels.
- To conduct assessments and evaluate student performance.

- To identify topics that require additional attention.
- To retrieve relevant educational videos.
- To provide personalized learning-resource recommendations.
- To monitor student learning progress.

5. Existing System

In conventional e-learning systems, course materials, quizzes, assessments and supplementary resources are often available through separate platforms.

Students may need to search manually for relevant videos and practice materials. They may also find it difficult to identify topics where they need additional learning support.

These limitations can make the learning process less organized and time-consuming.

6. Proposed System

EduPulse provides an integrated learning environment that combines course management, assessments, performance analysis and resource recommendations.

Students can access learning materials, attempt quizzes, view assessment results and identify topics that require improvement.

The platform integrates external services to generate quiz questions, retrieve educational videos and provide personalized learning-resource recommendations.

The proposed system supports a continuous learning process in which students study, practise, evaluate their performance and access additional resources.

7. Major Features

7.1 Student Login and Dashboard

Provides a student login interface and dashboard to access learning activities and personal learning information.

7.2 Course and Topic Management

Allows students to access available courses, select topics and explore lessons and learning materials.

7.3 Quiz Generation

Provides topic-based quizzes with Easy, Medium and Hard difficulty levels. Google Gemini API supports question generation, with a verified question bank available as a fallback.

7.4 Assessment and Evaluation

Allows students to attempt assessments and receive scores and performance information based on submitted answers.

7.5 Performance Analysis

Analyses assessment results to highlight topics where students may require additional practice.

7.6 Educational Video Retrieval

Uses YouTube Data API v3 to retrieve educational videos related to selected courses and topics.

7.7 Personalized Resource Recommendations

Integrates the StudyCompass recommendation service to provide learning resources based on student learning context and assessment performance.

7.8 Learning Progress Tracking

Maintains student learning information, assessment results and progress to support continuous learning.

8. System Architecture

EduPulse follows a client-server architecture with an external recommendation service. The architecture consists of the following components:

1 **Student:** Interacts with the platform by selecting courses, topics, difficulty levels and resource preferences.

2 **Frontend:** Provides the user interface using HTML, CSS, JavaScript, Jinja2 and Chart.js.

3 **Backend:** Uses Python FastAPI to handle application routes, quiz generation, assessment evaluation and recommendation requests.

4 **Database:** Uses SQLite with SQLAlchemy to store user details, course information, assessment results and learning progress.

5 **External APIs:** Integrates Google Gemini API, YouTube Data API v3 and StudyCompass API.

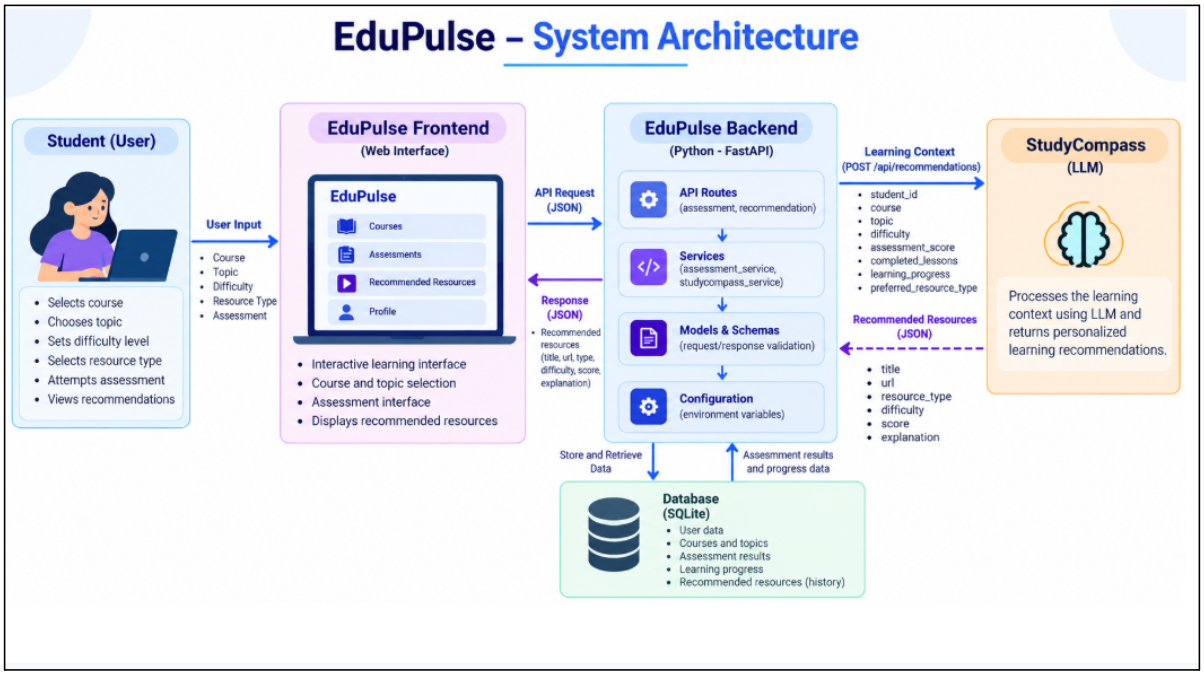


Figure 1: EduPulse System Architecture

9. Technology Stack

Component	Technology
Programming language	Python
Backend framework	FastAPI
Frontend	HTML, CSS, JavaScript
Template engine	Jinja2
Database	SQLite
ORM	SQLAlchemy
Data validation	Pydantic
Visualization	Chart.js
Quiz generation	Google Gemini API
Video retrieval	YouTube Data API v3
Recommendations	StudyCompass REST API
Deployment	Render
Version control	Git and GitHub

Table 1: Technologies used in EduPulse

10. API Integrations

10.1 Google Gemini API

Google Gemini API is integrated to support the generation of quiz questions based on the selected course, topic and difficulty level. Generated questions include multiple-choice options, correct answers and explanations. A verified question bank is used as a fallback when the API is unavailable or returns unsuitable results.

10.2 YouTube Data API v3

The YouTube Data API retrieves educational videos related to the selected course or topic. The retrieved information can include video titles, URLs, thumbnails and channel details.

10.3 StudyCompass API

StudyCompass is an external partner recommendation service integrated with EduPulse. The API receives learning-related information such as:

- Course
- Topic
- Difficulty level
- Assessment score
- Completed lessons
- Learning progress
- Preferred resource type

It processes the supplied information and returns personalized learning resources in JSON format. EduPulse displays the recommended resources to the student.

11. System Workflow

1 The student logs into the EduPulse platform.

2 The student selects a course and topic.

3 Learning materials and educational videos are accessed.

4 The student attempts quizzes or assessments.

5 The system evaluates answers and calculates the assessment score.

6 Student performance is analysed to identify topics requiring improvement.

7 The student selects the recommendation option.

8 EduPulse sends relevant learning information to the StudyCompass API.

9 StudyCompass processes the request and returns recommended resources.

10 The recommendations are displayed to the student for further learning.

12. Database Description

EduPulse uses SQLite as its database and SQLAlchemy for database operations. The database maintains information required for the platform, including:

- Student and user information
- Course and topic details
- Assessment questions and results
- Student learning progress
- Recommended resource history

The database helps maintain and retrieve student learning information throughout the learning process.

13. Results and Discussion

The EduPulse platform provides an integrated interface for accessing courses, learning materials, quizzes and assessments.

The quiz generation functionality supports the creation of topic-based questions with different difficulty levels. Students can attempt assessments and view their results.

The performance analysis functionality helps students identify topics that may require additional practice. Educational videos are retrieved through the YouTube Data API, while personalized learning-resource recommendations are obtained through the StudyCompass API.

The integration of these functionalities provides a connected learning experience within a single platform.

14. Advantages

- Provides an integrated learning environment.
- Supports quizzes with multiple difficulty levels.
- Helps students understand their assessment performance.
- Identifies topics requiring additional practice.
- Provides access to educational videos.
- Offers personalized learning-resource recommendations.
- Reduces the need to search across multiple platforms.
- Supports continuous learning and progress tracking.

15. Conclusion

EduPulse is a personalized e-learning platform that integrates course management, quiz generation, assessment evaluation, performance analysis and learning-resource recommendations.

By combining these functionalities with external API integrations, the platform provides students with a structured learning environment. It connects assessment performance with additional learning resources, helping students identify areas for improvement and continue their learning journey.

EduPulse demonstrates how web technologies and external services can be integrated to create a more organized and personalized digital learning experience.

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